Ayush Shukla

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EDUCATION

Manipal Institute of Technology, Manipal, India *Oct'20 – Jul'24*
Bachelor of Technology in **Electronics & Communication**

- *Cumulative GPA:* **7.62/10**
- *Bachelor's thesis*: "Designing of Surrogate models for Electromagnetic Compatibility of Power electronic circuits"

Field of Interest

- Radar Signal Processing, Frequency-Modulated Continuous-Wave radars, mmWave Radar Sensing, Target Detection and Estimation, Clutter Suppression, Sparse Reconstruction, Computational Electromagnetics, Antennas and RF systems, Surrogate Modelling, Unfolded Learning

PUBLICATIONS

- **A. Shukla**, S.S. Ram, "Low Rank and Sparsity Based Foreign Object Debris Detection with Millimeter Wave Radar", ***IEEE Radar Conference 2026***, Phoenix, Arizona, USA, 2026
- **A. Shukla**, S. S. Nukala, Akash, D. K. Singh, D. Gope, J. Hansen, "Comparison of Surrogate Modelling Approaches for Estimation of EMI Filter Insertion Loss", ***IEEE EDAPS 2023***, Mauritius, 2023

RESEARCH EXPERIENCE

Department of ECE, IIIT Delhi *May'25 – Present*
Research Associate | Supervisor: ***Dr. Shobha Sundar Ram,*** Professor, IIIT Delhi

- Working on **77 GHz Frequency-Modulated Continuous-Wave radar**-based **foreign object debris detection (FOD)** for airport runway safety using TI AWR1843BOOST radar platforms, in collaboration with the **Indian Air Force**
- Developed **signal-processing** pipelines for **low-Radar Cross Section** target detection (**~35 dBsm**) in ground clutter, including Binary phase modulation- MIMO, range-angle processing, background subtraction, radar calibration, and detection map generation
- Proposed a **low-rank** and **sparse decomposition** approach solved using **Alternating Direction Method of Multipliers (ADMM)** to separate structured ground clutter from sparse FOD targets in range-angle space, up to the range of **14** metres
- Conducted **RF anechoic chamber** and real-world **runway measurements** to evaluate the detection performance
- Extending research on the sparse optimization framework toward **learning-based unfolded methods** for more robust sensing
- This work resulted in a first-author paper accepted at **IEEE Radar Conference 2026**

Institute of Electronics, Graz University of Technology, Austria *Jan'24 – Jun'24*
Scientific Project Member | Supervisors ***- Dr. Jan Hansen,*** Assistant Professor, TU Graz

- Developed **Surrogate models** using Gaussian Processes for **Electromagnetic Compatibility** analysis of power electronic circuits
- **Reduced normalized prediction error** from 5.25% to 3.39% and cut training time by **33%** through algorithm enhancements
- Investigated the impact of **hyper-parameter tuning** and **sampling strategies** on model convergence and performance
- Benchmarked performance trade-offs leveraging ANSYS OptisLang DIM-GP framework, achieving a **high-fidelity model** for **Resonance-heavy regions**, and a 60% improvement in computational efficiency
- Conducted detailed model validation using the **Feature Selective Validation** (FSV) method in Python, incorporating Amplitude, Feature and Global difference measures across 6000-point signals, thereby bridging **numerical and qualitative evaluation**
- Documented and automated parts of the surrogate modeling workflow to support reuse in future EMC analysis workflows

Indian Institute of Science (IISc), Bangalore *Jun'23 – Jul'23*
Research Intern at the Numerics of Integrated Circuits and Electromagnetics (NICE) laboratory, Dept. of ECE
Supervisors ***- Dr. Jan Hansen,*** Visiting Professor, TU Graz, Austria| ***Dr. Dipanjan Gope,*** Associate Professor, ECE, IISc

- Analysed the trade off between prediction accuracy and training cost across surrogate modelling approaches for EMI filter insertion loss prediction
- Developed Surrogate models based on Kriging and **Polynomial Chaos Expansion** in UQLab with 15k+ training samples
- Implemented **Wideband Kriging** in MATLAB to achieve the highest accuracy with NRMSE ≈ 4.7% across 62 frequency samples
- Implemented Python-MATLAB interface, **Python-CST interface** for validating the machine learning model results
- This work resulted in a first-author paper accepted at **IEEE EDAPS Conference 2023**

Manipal Institute of Technology, Manipal *Feb'23 – Jun'23*
Research Intern | Supervisors – ***Dr. Tanweer Ali,*** Additional Professor, ECE, Manipal Institute of Technology

- Designed and simulated a **flexible Ultra-wideband** (UWB) antenna operating from 2.48 GHz to 11.91 GHz with a peak gain of 4.84 dB at 7.75 GHz using ANSYS HFSS for **Wireless body area network** (WBAN) applications
- Enhanced bandwidth and radiation characteristics through **parametric analysis** by introducing Cuts and reducing ground plane
- Validated mechanical robustness via **simulated bending tests** (R = 40/45 mm) with negligible impact on S11 and antenna gain

WORK EXPERIENCE

Robert Bosch, Bangalore *Aug'24 – April'25*

Assistant Manager

- Contributed to Product Strategy and Lifecycle Management (**PLCM**) for gasoline components in the Aftermarket channel, supporting a 2% year on year market share increase to **$75,000** through **data-driven portfolio planning**
- Led cross-functional coordination with sales, logistics, and global product teams in Germany for the introduction of Ignition coils, Fuel Supply modules into the Independent Aftermarket channel leading to a revenue stream of **$180,000**
- Engaged in site visits to Bosch manufacturing plants and service centers across India
- Participated as the youngest associate in the **annual strategic offsite retreat** to present and align the upcoming year's product strategy with senior management

PROJECTS

Review Report – Emerging Trends in 5G and Beyond *Sep'23 – Nov'23*

Course Project, 5G Fundamentals & Architectures

- Co-authored a technical review exploring the evolution from 5G to **6G**, focusing on **terahertz communication**, intelligent antennas, beamforming, and energy-efficient architectures
- Analysed key design challenges in **antenna miniaturization**, propagation at higher frequencies, and integration of AI/ML for adaptive and high-throughput wireless networks

Bluetooth Controlled Car *Sep'22 – Dec'22*

Course Project, Microprocessor Lab

- Developed a mobile-controlled vehicle using the **Intel 8051 microcontroller** and integrated **Bluetooth** communication for wireless navigation and control as part of a group

BCH Encoder & Decoder Simulation *Sep'23 – Nov'23*

Course Project, Error Control Coding

- Implemented a **BCH encoder and decoder** (15,7) using **NumPy**, simulating bit flips over a noisy channel
- Evaluated Bit Error Rate and basic error correction performance using **syndrome decoding**

Price Prediction *Apr'23 – May'23*

Course Project, Introduction to Data Analytics

- Developed a machine learning regression model in Python using **pandas** and **scikit-learn** to predict used car prices based on features like year, mileage, make, model, and engine power
- Performed **Exploratory Data Analysis** and model evaluation reporting performance metrics as **R^2 score** and **mean absolute error**

Audio Triggered Light Switch *Oct'21 – Dec'21*

Course Project, Digital System Design Lab

- Designed and implemented a **sound-activated** light control circuit on a breadboard using **IC 4017** and **op-amp 741**

SKILLS

Programming Languages and Tools: MATLAB, Python, C++, R, SQL, NumPy, Pandas, PyTorch, MS Office, LATEX
Software: mmavestudio, ANSYS HFSS, CST Studio, UQLab, OptisLang, LTSpice
Radar systems: TI AWR1843BOOST, AWR2243BOOST
Languages: English, Hindi

RELEVANT COURSEWORK

Electronics Engineering: Basic Electrical Technology (*), Basic Electronics (*), Analog Electronic Circuits (*), Network Analysis (*), Electromagnetic Waves, Digital Signal Processing, Microwave Engineering, Microprocessors, Communication Networks, Wireless Communication, Error Control Coding, 5G Fundamentals and Architectures, Linear Control Theory
Computer Science: Data structures and Algorithms, Computer Architecture, Fundamentals of Computing (*), Data Analytics, Data Science (*), Principles of Soft Computing
Mathematics: Linear Algebra, Partial Differential Equations (*), Complex Analysis, Probability and Distribution
Other: Engineering Physics (*), Engineering Economics, Essentials of Management (*), Mechanics of Solids

Received A grade in courses marked with (*)

EXTRACURRICULAR ACTIVITY

- Played Drums and performed in the cultural segment of the annual Bosch retreat
- Led external delegate coordination for Revels, the **annual cultural and sports fest** of Manipal Institute of Technology, facilitating smooth registration and travel for **3000+ students nationwide**; received **Dean's appreciation** in the year 2023
- Volunteered with Project Mudra at MIT Manipal in **teaching elementary mathematics** to underprivileged children
- Senior team member in Rugved Systems, a student-led project on autonomous defense robotics; **managed budgets and recruitment** in the year 2022